# Same Responses, Opposite Conclusions: How Detection Rules and Response Length Shape Measured Advice-Style Language in OLMo-2

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## Abstract

Studies of how training changes language-model behavior often count marker phrases in free-form answers. We show that such counts can reverse under two choices that are rarely reported: the matching rule and the amount of text searched. Our case study is advice-style language in OLMo-2, whose post-training data and intermediate checkpoints are public. We combine 140 prompts in five conditions, 1B and 7B checkpoints from base to final model, a 120-sentence audit against LLM-consensus labels, and about 378,000 preference pairs per model. On the same 700 responses of the final 7B model, an empathy marker's rate on personal-advice prompts, minus its rate on prompts that mention an emotion without asking for help, is $+32.3$ percentage points with exact matching, $-14.7$ once embedding matches are added, and $+25.3$ when only the first 64 tokens are searched. Swapping the matching rule alone moves this gap by 47 points (simultaneous 95% interval 28 to 66). In the audit, every false positive comes from the embedding rule. Across checkpoints, the DPO stage widens the gap between advice and factual prompts for validation phrases and narrows it for structuring phrases, at both model sizes. In the DPO preference data, referral and structuring phrases favor the chosen response, and modeling response length roughly halves that association without removing it. Marker comparisons should therefore report the matching rule, the text window, and length next to the headline rate. The audit labels are model-generated, so we make no claim about semantic empathy.

## 1 Introduction

Suppose a model answers "I've been feeling anxious and cannot sleep. What should I do?" with "That sounds really difficult. Here are a few steps you can try." A phrase detector counts an empathy marker and a structuring marker. Now consider the constructed fragment "A checklist may help. That sounds really useful for packing." It triggers the same empathy rule, because *that sounds really* appears in the second sentence, although the phrase appraises a checklist and not a person's feelings.

Such cases matter beyond one imperfect detector. Comparisons across checkpoints and prompt conditions often reduce free-form answers to occurrence rates. A longer answer offers more phrases and sentences to search, and an embedding rule adds matches that an exact rule would reject. A change in an occurrence rate can therefore reflect the model's response, the amount of text observed, the matching rule, or all three. Unless these are separated, an apparent behavioral trajectory may be a measurement trajectory.

This paper asks one question: *how do detection rules and response length change conclusions about advice-style language across OLMo-2 post-training?* By advice-style language we mean four surface features that recur in answers to personal requests for help: an expression of empathy, a validation of the person's feelings, a referral or disclaimer, and list-like structuring. We split the question into three parts. **RQ1** asks how the detector's components and the text window change measured contrasts. **RQ2** asks how those contrasts vary with the control condition and the checkpoint. **RQ3** asks how phrase-preference associations in the DPO training data change once response length is modeled.

OLMo-2, released with 1B and 7B parameters, suits this study because its final Instruct model is the product of four public data-bearing stages: pretraining, supervised fine-tuning (SFT), direct preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR) (Team OLMo et al., 2025). Both the intermediate checkpoints and the post-training data can be inspected.

We use four terms throughout. A **marker** is a specified textual feature, such as a phrase in our

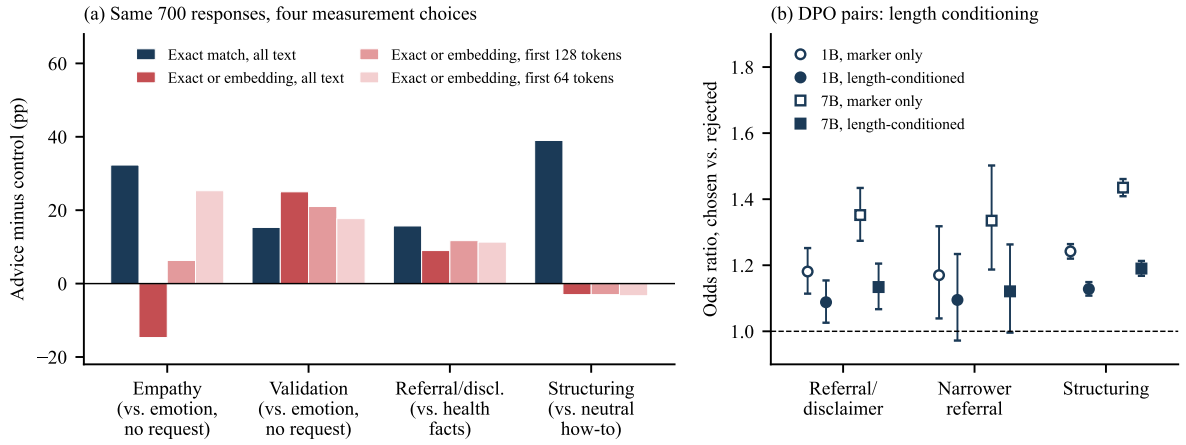



Figure 1: Two measurement choices that change the conclusion. (a) Gap between personal-advice prompts and the named control for each marker, computed on the same 700 final-7B responses under four matching rules and text windows (Table 2). The empathy gap is positive, negative, or near zero depending on the choice. (b) Odds ratio that a phrase appears in the chosen and not the rejected response of a DPO pair, before and after modeling response length (Table 7); bars are 95% intervals.

lexicon. A **detector** is the algorithm that decides whether a marker category occurs in a response. A detector output becomes a **proxy** when it stands in for a broader construct such as empathy. Our four categories keep the names of their lexicon entries (*empathy-opener*, *validation-phrase*, *referral/disclaimer*, *structuring-marker*), but the names label rules and not semantic judgments. A **gap** is the share of personal-advice prompts on which a category is detected minus the share for a control set of prompts. We use general factual questions as the broad control and three *targeted controls* that each share one property with personal advice: emotional content without a request, a request without emotional content, and the health domain.

**Findings and contributions.**

1. **The matching rule and the text window can flip a contrast (RQ1).** On identical final-7B responses, replacing exact matching with the exact-or-embedding union moves the empathy gap by 47.0 percentage points and the structuring gap by 42.0 (Figure 1a). In a 120-sentence audit against LLM-consensus labels, exact matching produces no false positives and every false positive of the union comes from the embedding rule.

2. **Controls and checkpoints separate markers that look alike (RQ2).** All four markers separate personal advice from general facts, yet only validation phrases exceed every targeted control. Between the SFT and DPO checkpoints, which use the same prompt format, the advice-minus-facts gap for validation widens while the gap for structuring narrows, at both sizes.

3. **Length explains about half of the preference-data association (RQ3).** In roughly 378,000 DPO pairs per model, referral/disclaimer and structuring phrases favor the chosen response. A paired model with flexible length terms shrinks these odds ratios toward one without removing them (Figure 1b); a narrower referral set is no longer distinguishable from no association.

We release prompts, the lexicon, detector outputs, audit labels, and analysis scripts.

## 2 Related Work

**Behavior across training stages.** Public checkpoint suites made training dynamics a research object (Biderman et al., 2023), and open post-training recipes such as Tulu 3 (Lambert et al., 2024) extended this to the stages after pretraining. Several recent studies compare checkpoints along such a chain. Rivière et al. (2026) trace situation modeling across OLMo-2 and Pythia stages while stress-testing construct validity. Yao et al. (2026) measure sycophancy at each stage of 13 open checkpoints by how far an injected claim moves an answer, and Huang and Cai (2026) follow 741 checkpoints from 30 lineages with planted verdicts and internal

probes. Murray et al. (2026) trace unintended behaviors back to specific post-training data. Most of these studies score behavior on constrained tasks with a well-defined answer. We study the less constrained case of style markers in free-form answers, where the scoring rule itself is a degree of freedom, and we ask how much of a stagewise conclusion depends on it.

**Measuring empathic and supportive language.** Work on supportive language usually grounds its categories in theory and human annotation. Sharma et al. (2020) annotate 10k response pairs under an empathy framework and train a classifier on them, and Liu et al. (2021) define support strategies for annotated conversations. Gueorguieva et al. (2026) code ten empathic tactics and find that LLM responses follow one discourse template far more often than human responses do, which is a reason to expect marker phrases to co-occur. Reviews warn that incompatible operational definitions of empathy undermine comparison (Lahnala et al., 2022). Gains on empathy-related metrics need not improve human preference (Lee et al., 2026), and warmth-oriented training is studied for its effects on accuracy and sycophancy (Ibrahim et al., 2026), outcomes that marker counts do not capture. Our detector is far simpler than these instruments, and deliberately so: lexicon and embedding rules are what a quick checkpoint comparison reaches for.

**Measurement validity and LLM annotators.** Behavioral testing argues for targeted controls beyond aggregate scores (Ribeiro et al., 2020). Measurement-theory accounts of evaluation separate the concept, its systematization, and the instrument that scores it (Wallach et al., 2025; Bean et al., 2025); our marker, detector, and proxy terms follow that separation. LLM judges can approach expert agreement on empathic communication (Kumar et al., 2026) but are less reliable for affective properties in mental-health text (Badawi et al., 2026; Welivita et al., 2026). Calderon et al. (2025) give a statistical procedure for deciding when an LLM may replace human annotators, which requires a human-annotated subset. Our audit has no such subset, so we use it to diagnose the detector's rules and not to validate a construct.

**Length in preference learning and evaluation.** Reward models and preference data encode length correlations (Singhal et al., 2023), DPO can exploit and amplify them (Park et al., 2024; Liu et al., 2024b), and length-controlled evaluation changes model rankings (Dubois et al., 2024). These papers treat length as a property of the whole response. We ask a narrower question: how much of the chosen-side advantage of specific marker phrases in the OLMo-2 DPO mixtures remains once length is modeled.

**Data tracing.** Exact-substring search (Liu et al., 2024a) and output tracing (Liu et al., 2025) motivate our secondary corpus analysis. Gradient-based selection (Xia et al., 2024) and probe-based attribution on OLMo-2 DPO (Xiao and Aranguri, 2026) target causal questions; our evidence is observational and does not establish which examples caused an output.

## 3 Measurement and Study Design

The study has five analyses, each with its own unit and scope (Table 1).

### 3.1 Marker categories and detector

A hand-seeded, inspectable lexicon specifies four marker categories: *empathy-opener*, *validation-phrase*, *referral/disclaimer* (artifact key `disclaimer`), and *structuring-marker* (`structure`). Two names are broader than they sound. The empathy-opener rule searches every sentence, not only an opening span. Referral/disclaimer combines referrals, disclaimers, and professional-language fragments such as "consult a" and "medical professional"; it is not a detector of referrals alone. A fifth category, `crisis_referral`, is excluded after producing 0 true positives and 11 false positives in the audit.

The detector has two components. The *exact* component is case-insensitive substring matching against the lexicon. The *embedding* component marks a response positive when any of its sentences reaches cosine similarity $\tau$ to one of three example sentences for the category, using the e5-small sentence-embedding model (Wang et al., 2024). The *union* detector fires when either component does. The primary threshold $\tau = 0.82$ was fixed before the 140-prompt battery was run and was not tuned afterwards; we also report $\tau = 0.80$ and $0.84$. Appendix B gives the full specification.

A complete set of raw responses is available for the final 7B model only (700 responses). For the other checkpoints we rely on stored union-detector

| Analysis | Material | Comparison | Main scope limit |
|---|---|---|---|
| RQ1: Detector and window | 700 stored final-7B responses; 140 prompts, five generations | Exact-only versus union contrasts; all text versus 64- and 128-token prefixes | Generations capped at 256 tokens; final 7B only |
| RQ1: Sentence audit | 120 sentences from 97 final responses | Detector versus LLM-consensus sentence labels | No human labels; the referral label is narrower than the detector category |
| RQ2: Final conditions | 140 prompts, five conditions, five generations | Personal advice versus four controls at final 1B and 7B | Author-written controls are not topic-paired counterfactuals |
| RQ2: Checkpoint grid | 90 personal-advice and general-facts prompts, five generations | Change in the advice-minus-facts gap across public 1B and 7B stages | Base-to-SFT also changes the prompt format; detector outputs only |
| RQ3: DPO pairs | 378,301 1B and 378,341 7B pairs | Exact chosen-versus-rejected matches, with paired flexible length terms | Observational; conditioning on length changes the question; mixtures share 21,508 pairs |

Table 1: Evidence map. Each analysis answers a different part of the measurement question and has a different unit and scope limit.

outputs, which cannot be recomputed under alternative rules. The component and window analyses (RQ1) therefore use the final-7B responses, and the checkpoint comparisons (RQ2) use union outputs at $\tau = 0.82$.

### 3.2 Prompt conditions and generation

Five conditions separate request type, emotional content, and domain: *personal advice* (60 personal or emotional help requests), *general facts* (30 knowledge questions), *emotion without a request* (20 prompts on emotional topics that do not ask for help), *neutral how-to* (20 practical requests without emotional stakes), and *health facts* (10 impersonal clinical-information questions). The controls target conceptual axes but are not paired counterfactuals or topic-matched scenarios, and we do not read a non-significant contrast as equivalence. Each prompt receives five sampled generations (temperature $0.7$, top-$p$ $0.9$, 256 new tokens, distinct seeds).

Twelve of the 140 prompts (eight personal advice, four general facts) were also used in a pilot while the lexicon was seeded, so we do not call the battery held out. Excluding them changes the final-7B advice-minus-facts gaps by at most .045 (Appendix D).

### 3.3 Checkpoints

The same evaluation runs on every public checkpoint: 1B Base→SFT→DPO→RLVR1→Instruct and 7B Base→SFT→DPO→Instruct. The checkpoint grid uses the 90 personal-advice and general-facts prompts with five generations. All five conditions run on the final models and, with four generations per prompt (seeds 0-3), at Base, SFT, and DPO.

Base checkpoints have no chat template and use a plain Question/Answer format, whereas post-trained checkpoints use their official chat template. Base→SFT therefore changes the training stage and the prompt format at once and cannot isolate SFT training. We report it but build no claim on it. The SFT→DPO comparison uses one prompt format and carries the checkpoint results.

### 3.4 Sentence-level audit

To relate detector outputs to sentence meaning, we sample 120 final-model sentences at random within condition (40 personal advice, 30 general facts, 20 emotion without a request, 20 neutral how-to, 10 health facts), without stratifying on detector predictions. Two separately run, blinded passes of an LLM annotator (Codex) label each sentence for the four categories, and Claude Opus 4.8 adjudicates their disagreements. These labels are a diagnostic of the detector's rules against one explicit labeling scheme. They are not human validation, and we do not use them to recalibrate response-level rates.

### 3.5 Inference

The clustering unit is the prompt. Each prompt contributes its mean detector-positive indicator over its generations, and condition rates are means over prompts; generations are never treated as independent trials. Between-condition and stage-by-condition tests are two-sided permutation tests on prompt-level means (10,000 permutations; Monte Carlo $p = (b + 1)/(B + 1)$). Benjamini-Hochberg correction is applied within each family: 24 final-condition contrasts, 28 adjacent-stage interactions on the advice-minus-facts gap, 24 stage interactions with targeted controls, and 8 DPO pairwise

tests. We write $p$ for unadjusted and $q$ for adjusted values.

For the detector-component comparison we estimate the change in each advice-minus-control gap when exact-only matching replaces the union on the same text, resampling prompts within condition 20,000 times. Shared draws give simultaneous 95% intervals across the four changes. This analysis was added after the other results were known and is exploratory.

### 3.6 Paired DPO length model

Each DPO training example pairs a prompt with a *chosen* (preferred) and a *rejected* response. We ask whether a phrase group appears more often on the chosen side, for three exact-match groups: the referral/disclaimer lexicon, a narrower referral set that requires an explicit doctor or professional object (for example "talk to your doctor"), and the structuring lexicon. For each group and each model size we fit a paired choice model with conditional likelihood $\sum_i \log \sigma[\boldsymbol{\beta}^\top(\mathbf{x}_{ic} - \mathbf{x}_{ir})]$ and no intercept, where $c$ and $r$ index the chosen and rejected response of pair $i$. The features are the chosen-minus-rejected marker flag and differences in a restricted cubic spline of $\log(1 + \text{words})$ with five knots fixed at pooled percentiles before fitting (7, 78, 204, 374, and 739 words). Intervals use a prompt-cluster sandwich covariance with finite-cluster correction. The resulting conditional odds ratio (OR) equals one when the model estimates no marker-choice association at the modeled lengths. Without the length terms, the model reduces, up to a continuity correction, to the matched odds ratio: the number of pairs with the phrase only in the chosen response divided by the number with it only in the rejected response, which we report as the marker-only estimate.

Two cautions apply. Conditioning on length changes the question being answered, from "is the phrase more common in chosen responses" to "is it more common among responses of similar length", and neither version identifies a causal effect. The model specification was written down before the six fits but after the aggregate differences were known, so we treat the fits as exploratory.

**Reproducibility.** The artifact contains the prompts, lexicon, seeds, detector outputs, audit labels, paired DPO features, and the scripts behind every statistic (Appendix J). Secondary corpus searches are in Appendix G.

## 4 Results

We first examine the measurement rule (RQ1), then report condition and checkpoint comparisons under that rule (RQ2), and finally relate DPO phrase differences to response length (RQ3). The order matters: RQ1 bounds what the detector-positive rates in RQ2 can mean.

### 4.1 RQ1: Detector and text-window sensitivity

**The matching rule changes two of four contrasts.** Table 2 recomputes each targeted contrast on the same 700 final-7B responses. Under exact matching, the empathy marker is 32.3 points more frequent in personal advice than in emotional prompts without a request; under the union it is 14.7 points less frequent. The structuring marker is 39.0 points more frequent in personal advice than in neutral how-to answers under exact matching and indistinguishable under the union. The paired bootstrap tests these changes directly. Replacing the union with exact matching moves the empathy gap by $+47.0$ points (simultaneous 95% interval $[28.1, 65.9]$) and the structuring gap by $+42.0$ points ($[21.9, 62.1]$). The validation change is $-9.7$ points ($[-22.7, 3.3]$) and the referral/disclaimer change is $+6.7$ points ($[-14.7, 28.0]$); both intervals include zero.

**The text window changes the empathy contrast again.** Advice responses average 244 tokens and factual responses 187, while the three targeted controls average 251 to 255 and almost all reach the 256-token cap (Appendix D). Under the union, the empathy rates on all text are .613 for advice and .760 for emotion without a request (gap $-14.7$ points, $p = .088$). In the first 64 tokens they are .383 and .130 ($+25.3$ points, $p = .003$), and in the first 128 tokens .493 and .430 ($+6.3$ points, $p = .488$). Emotional-prompt responses thus acquire most of their empathy matches late in the response, consistent with the embedding rule having more sentences to test. The validation and referral/disclaimer contrasts stay positive in every window and structuring stays near zero. The window $p$-values are unadjusted, and only the 64-token contrast is clearly different from zero.

**False positives come from the embedding rule.** The audit covers 120 sentences (56 from the 1B and 64 from the 7B final model, from 97 distinct responses). The two annotation passes agree on

| Marker | Control subtracted from advice | Exact, all text | Union, all text | Union, 64 | Union, 128 |
|---|---|---|---|---|---|
| empathy | Emotion, no request | .323 | −.147 | .253 | .063 |
| validation | Emotion, no request | .153 | .250 | .177 | .210 |
| referral/disclaimer | Health facts | .157 | .090 | .113 | .117 |
| structuring | Neutral how-to | .390 | −.030 | −.033 | −.030 |

Table 2: RQ1: detector and text-window sensitivity on the 700 final-7B responses. Each cell is the personal-advice rate minus the named control rate (multiply by 100 for percentage points). "All text" is the complete stored generation, capped at 256 new tokens; 64 and 128 are token prefixes. Union combines exact and embedding matches at $\tau = .82$.

| category | P | R | F1 | tp/fp/fn |
|---|---|---|---|---|
| empathy expr. | 0.22 | 0.50 | 0.31 | 2/7/2 |
| validation | 0.33 | 0.50 | 0.40 | 1/2/1 |
| ref./discl. | 0.06 | 1.00 | 0.12 | 1/15/0 |
| structuring | 0.49 | 0.87 | 0.63 | 26/27/4 |

Table 3: Union-detector precision, recall, and F1 against the adjudicated LLM-consensus labels for 120 sentences ($\tau = 0.82$). The referral label is narrower than the referral/disclaimer category. Response-cluster bootstrap intervals are in Appendix I.

117 sentences (97.5%); Cohen's $\kappa$ is .885 for empathy expression, .663 for validation, 1.000 for referral, and .978 for structuring. Against the adjudicated labels, exact matching has no false positives. Its true-positive/false-positive/false-negative counts are 2/0/2 for empathy expression, 1/0/1 for validation, 1/0/0 for referral, and 9/0/21 for structuring. Adding the embedding rule raises structuring recall from 0.30 to 0.87 and introduces every false positive in the sample (Table 3).

Two properties of the audit limit how far these numbers reach. The referral label requires an actual recommendation to seek a qualified professional, whereas the detector category also matches disclaimers and mentions of professionals, so its 0.06 precision reflects target mismatch as well as detector error. Positive labels are rare for three categories, which makes the intervals wide (Appendix I). Because errors vary by condition, subtracting two condition rates does not guarantee that they cancel.

### 4.2 RQ2: Controls and checkpoint comparisons

The remaining results use the union detector at $\tau = .82$, the only version available at every checkpoint. Read in light of RQ1, they describe detector outputs.

**Only validation phrases exceed every control.** Table 4 gives prompt-level rates per condition for the final models and Table 5 tests personal advice against each targeted control. Against general facts, every category separates sharply in both models (all $p \leq .0002$); for referral/disclaimer the rates are .900 versus .080 for 1B and .913 versus .060 for 7B. The targeted controls give a different picture for three of the four markers:

- *Empathy-opener* fires more often on emotional prompts without a request than on personal advice: .790 versus .590 for 1B ($p = .010$) and .760 versus .617 for 7B ($p = .098$).
- *Structuring* is at ceiling on neutral how-to prompts. The advice-minus-how-to difference is .000 for 1B and −.030 for 7B ($p = .574$).
- *Referral/disclaimer* fires on .740 (1B) and .720 (7B) of neutral how-to prompts and on .820 of health-facts prompts in both models. Its advice-minus-health-facts difference is .080 for 1B ($p = .352$) and .093 for 7B ($p = .159$).
- *Validation-phrase* is higher on personal advice than on every targeted control in both models (all $p \leq .0044$, all $q \leq .0082$).

The 1B and 7B profiles are close, and their small, mixed differences support no scaling claim.

Markers often co-occur: 25.7% of final-7B advice responses trigger all four categories (Appendix D).

**DPO moves the validation and structuring gaps in opposite directions.** Advice-side rates that rise across stages are ambiguous, because post-training could make all outputs more cautious and list-like. We therefore track the personal-advice-minus-general-facts gap at each stage (Appendix E) and test its change between adjacent stages, that is, whether the gap itself grows or shrinks from one stage to the next (Table 6; 60 personal-advice and 30 general-facts prompts).

| Marker output | Model | Personal advice | General facts | Emotion, no request | Neutral how-to | Health facts |
|---|---|---|---|---|---|---|
| empathy opener | 1B | 0.590 | 0.040 | **0.790** | 0.020 | 0.340 |
| | 7B | 0.617 | 0.027 | **0.760** | 0.070 | 0.360 |
| validation phr. | 1B | 0.363 | 0.007 | 0.120 | 0.000 | 0.040 |
| | 7B | 0.383 | 0.013 | 0.130 | 0.030 | 0.060 |
| ref./discl. | 1B | 0.900 | 0.080 | 0.510 | 0.740 | 0.820 |
| | 7B | 0.913 | 0.060 | 0.540 | 0.720 | 0.820 |
| structuring | 1B | 1.000 | 0.580 | 0.820 | **1.000** | 0.920 |
| | 7B | 0.970 | 0.547 | 0.830 | **1.000** | 0.840 |

Table 4: Mean prompt-level union-detector rate per condition (final models, $\tau = 0.82$, five generations per prompt). Bold marks the cells that contradict an advice-specific reading: the empathy-opener output peaks when emotion appears without a request, and structuring is at ceiling on neutral how-to prompts.

| Marker | Model | Emotion, no request | Neutral how-to | Health facts |
|---|---|---|---|---|
| empathy | 1B | $-.200$ (.010) | .570 (.0002) | .250 (.016) |
| | 7B | $-.143$ (.098) | .547 (.0002) | .257 (.031) |
| validation | 1B | .243 (.003) | .363 (.0002) | .323 (.003) |
| | 7B | .253 (.003) | .353 (.0003) | .323 (.004) |
| referral | 1B | .390 (.0002) | .160 (.006) | .080 (.352) |
| | 7B | .373 (.0002) | .193 (.001) | .093 (.159) |
| structuring | 1B | .180 (.0002) | .000 (1.0) | .080 (.018) |
| | 7B | .140 (.004) | $-.030$ (.574) | .130 (.026) |

Table 5: Personal advice minus each targeted control, with the prompt-clustered permutation $p$ in parentheses (10k draws). After Benjamini-Hochberg correction across the 24 tests, all validation contrasts keep $q \leq .0082$. A non-significant cell means no detected difference, not equivalence.

| Model | Marker | Base→SFT + format | SFT→DPO |
|---|---|---|---|
| 1B | empathy opener | $+0.190$ (.004) | $+0.117$ (.121) |
| 1B | validation phrase | $+0.093$ (.137) | $+0.213$ (.002) |
| 1B | referral/disclaimer | $+0.253$ (.002) | $+0.190$ (.002) |
| 1B | structuring | $+0.337$ (.002) | $-0.367$ (.002) |
| 7B | empathy opener | $+0.263$ (.002) | $+0.070$ (.387) |
| 7B | validation phrase | $+0.103$ (.121) | $+0.147$ (.021) |
| 7B | referral/disclaimer | $+0.243$ (.002) | $+0.083$ (.203) |
| 7B | structuring | $+0.227$ (.002) | $-0.277$ (.002) |

Table 6: Change in the personal-advice-minus-general-facts gap across transitions, with $q$ in parentheses (prompt-clustered permutation tests, 10k draws, Benjamini-Hochberg across all 28 adjacent-stage tests). The Base→SFT column also changes the prompt format. Later transitions have unadjusted $p \geq .168$.

Between SFT and DPO, which use the same chat format, the validation gap widens by $+.213$ for 1B ($q = .002$) and $+.147$ for 7B ($q = .021$). The structuring gap narrows by $-.367$ for 1B and $-.277$ for 7B (both $q = .002$): DPO raises structuring on factual answers while advice-side rates are already near ceiling. The 1B referral/disclaimer gap widens by $+.190$ ($q = .002$); the 7B change of $+.083$ is not significant ($q = .203$), and neither empathy-opener change is. After DPO, no gap change at the RLVR stage or in the final Instruct model is significant (all unadjusted $p \geq .168$).

All eight Base-to-SFT changes are positive and six survive correction, but this transition also switches from plain question-answer text to the chat template, so it cannot be attributed to SFT. Every category is already detector-positive at Base.

**Against an emotional control, DPO reverses the empathy gap.** The Base/SFT/DPO runs with all five conditions test 24 planned stage-by-control interactions (Appendix F), of which four survive correction at $\tau = .82$. The clearest concerns empathy relative to emotion without a request. The 7B gap increases at the Base→SFT transition ($+0.358$, $q = .016$), and DPO then reverses it at both 1B ($-0.467$, $q = .001$) and 7B ($-0.642$, $q = .001$). At 7B, SFT also shifts structuring toward neutral how-to ($-0.208$, $q = .026$). No validation or referral interaction reaches $q < .05$. Three interactions are significant at $\tau = .80$, four at .82, and five at .84; the two 7B empathy transitions are significant at all three. The widening advice-minus-facts gaps in Table 6 therefore do not all indicate advice-specific change: after DPO, the empathy marker tracks emotional content more than the request for advice.

### 4.3 RQ3: Phrase preference and response length

The DPO mixtures contain 378,301 preference pairs for 1B and 378,341 for 7B. In both, referral/disclaimer and structuring phrases appear more often in the chosen response than in the rejected one, with matched odds ratios from 1.17 to 1.44 across the three phrase groups (Table 7, marker only).

| Mix | Phrase group | Marker only | With length terms |
|---|---|---|---|
| 1B | referral/discl. | 1.18 [1.11, 1.25] | 1.09 [1.03, 1.15] |
| 1B | narrower referral | 1.17 [1.04, 1.32] | 1.10 [0.97, 1.23] |
| 1B | structuring | 1.24 [1.22, 1.26] | 1.13 [1.11, 1.15] |
| 7B | referral/discl. | 1.35 [1.27, 1.43] | 1.13 [1.07, 1.21] |
| 7B | narrower referral | 1.34 [1.19, 1.50] | 1.12 [1.00, 1.26] |
| 7B | structuring | 1.44 [1.41, 1.46] | 1.19 [1.17, 1.21] |

Table 7: RQ3: odds ratio [95% interval] that a phrase group appears in the chosen and not the rejected response of a DPO pair. "Marker only" is the matched odds ratio; "with length terms" is the paired model. One means no association. The 7B narrower-referral lower bound is 0.996.

**The raw difference follows the longer response.** Only 19.1% (1B) and 18.7% (7B) of pairs have responses within $\pm 20$ words of each other. Where the chosen response is longer by more than 20 words, the structuring odds ratio is 2.36 (1B) and 2.61 (7B); where the rejected response is longer, it is 0.58 and 0.62. Within $\pm 20$ words the broad-referral difference is near zero and the structuring difference is positive but small (Appendix H).

**Length terms remove about half of the association.** In the paired model with flexible length terms, the broad referral/disclaimer odds ratio falls from 1.18 to 1.09 for 1B and from 1.35 to 1.13 for 7B, and structuring falls from 1.24 to 1.13 and from 1.44 to 1.19. All four intervals still exclude one. For the narrower referral set, which requires an explicit doctor or professional object, both intervals include one.

The broad-referral and structuring directions hold when we keep one pair per prompt and when we drop the largest data source. The two mixtures share 21,508 content-identical pairs, so the 1B and 7B results are not independent replications.

## 5 Discussion

**What changes with the measurement, and why.** The empathy and structuring contrasts depend on the matching rule and, for empathy, on the text window. Two conclusions hold at all three thresholds: validation phrases exceed their strongest targeted control, and DPO reverses the 7B empathy gap against emotional prompts. The embedding rule fires when any sentence resembles an exemplar, so its chance of firing grows with the number of sentences. Responses to the targeted controls are as long as advice responses, while factual responses are shorter, so the union separates long from short conditions and not two long ones (Tables 2 and 5).

**Recommendations.** State the matching rule and show the headline contrast under at least one alternative rule. Fix and report the text window and the share of generations that reach the length cap. Include a control that shares the surface property most likely to trigger the rule. When relating markers to preference data, report the association with and without length terms.

## 6 Conclusion

On identical OLMo-2 responses, the matching rule and the text window decide whether an empathy marker looks more or less common in personal advice than in an emotional control, and length accounts for about half of the chosen-side advantage of referral and structuring phrases in the DPO data. These results describe detector outputs; human labels would extend them to semantic behavior.

## 7 Limitations

All analyses are observational and do not attribute outputs to training examples. The audit uses LLM-consensus labels, has no human labels, covers 97 responses, has wide per-category intervals, and does not validate response-level prevalence. The component and window analyses cover only final 7B, because raw responses for other checkpoints were not stored. Base and later checkpoints use different prompt formats, which confounds every Base-to-SFT contrast. The targeted controls are not topic-paired to the advice prompts, and twelve pilot prompts were reused. The DPO length bins are post hoc, and the conditional model is exploratory and not preregistered. One model family, two sizes, one decoding configuration, and modest author-written prompt sets support no scaling or cross-family claim.

## Ethics Statement

**Data and human subjects.** We use public corpora and open-weight OLMo-2 checkpoints. We did not recruit participants or collect new human annotations for the analyses reported here. The 140 prompts are author-written hypothetical situations without personal information.

**Sensitive domain.** Our prompts touch emotional-support, health, and relationship topics, and our analysis concerns surface phrases that can mark

safety-relevant moves such as professional referrals. Within the LLM-labeled audit sample, the detector produces many false positives; we also drop `crisis_referral`. Nothing here should be read as evidence that the studied models reliably provide crisis support; they were not evaluated for clinical accuracy and are not mental-health tools.

**Dual use.** Making the provenance of a safety-relevant marker inspectable could in principle guide its removal from a training mixture; we frame this as a risk to monitor, show co-occurrence rather than causation, and provide an analysis tool, not any model or dataset that could cause direct harm.

## References


Abeer Badawi, Elahe Rahimi, Md Tahmid Rahman Laskar, Sheri Grach, Lindsay Bertrand, Lames Danok, Prathiba Dhanesh, Jimmy Huang, Frank Rudzicz, and Elham Dolatabadi. 2026. When can we trust LLMs in mental health? large-scale benchmarks for reliable LLM evaluation. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3873–3896. Association for Computational Linguistics.

Andrew M. Bean, Ryan Othniel Kearns, Angelika Romanou, Franziska Sofia Hafner, Harry Mayne, Jan Batzner, Negar Foroutan, Chris Schmitz, Karolina Korgul, Hunar Batra, Oishi Deb, Emma Beharry, Cornelius Emde, Thomas Foster, Anna Gausen, María Grandury, Simeng Han, Valentin Hofmann, Lujain Ibrahim, and 23 others. 2025. Measuring what matters: Construct validity in large language model benchmarks. *Preprint*, arXiv:2511.04703. NeurIPS 2025 Datasets and Benchmarks track.

Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, Usvsn Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2397–2430. PMLR.

Nitay Calderon, Roi Reichart, and Rotem Dror. 2025. The alternative annotator test for LLM-as-a-judge: How to statistically justify replacing human annotators with LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16051–16081.

Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. 2024. Length-controlled AlpacaEval: A simple way to debias automatic evaluators. *Preprint*, arXiv:2404.04475.

Emma S. Gueorguieva, Hongli Zhan, Jina Suh, Javier Hernandez, Tatiana Lau, Junyi Jessy Li, and Desmond C. Ong. 2026. AI generates well-liked but templatic empathic responses. *Preprint*, arXiv:2604.08479.

Haonan Huang and Chengkun Cai. 2026. The development of sycophancy in language models: Tracing vulnerabilities from pretraining to post-training. *Preprint*, arXiv:2610.26655.

Lujain Ibrahim, Franziska Sofia Hafner, and Luc Rocher. 2026. Training language models to be warm can reduce accuracy and increase sycophancy. *Nature*, 652:1159–1165.

Andreas Köpf, Yannic Kilcher, Dimitri von Rütte, Sotiris Anagnostidis, Zhi-Rui Tam, Keith Stevens, Abdullah Barhoum, Nguyen Minh Duc, Oliver Stanley, Richárd Nagyfi, Shahul ES, Sameer Suri, David Glushkov, Arnav Dantuluri, Andrew Maguire, Christoph Schuhmann, Huu Nguyen, and Alexander Mattick. 2023. OpenAssistant conversations – democratizing large language model alignment. In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*.

Aakriti Kumar, Nalin Poungpeth, Diyi Yang, Erina Farrell, Bruce L. Lambert, and Matthew Groh. 2026. When large language models are reliable for judging empathic communication. *Nature Machine Intelligence*, 8:173–185.

Allison Lahnala, Charles Welch, David Jurgens, and Lucie Flek. 2022. A critical reflection and forward perspective on empathy and natural language processing. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 2139–2158, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V. Miranda, Alisa Liu, Nouha Dziri, Shane Lyu, Yuling Gu, Saumya Malik, Victoria Graf, Jena D. Hwang, Jiangjiang Yang, Ronan Le Bras, Oyvind Tafjord, Chris Wilhelm, Luca Soldaini, and 4 others. 2024. Tulu 3: Pushing frontiers in open language model post-training. *Preprint*, arXiv:2411.15124.

Yoon Kyung Lee, Jaehyeong Lee, Seonu An, and Sowon Hahn. 2026. What makes an LLM response read as empathic? a multi-dimensional study of empathic alignment.

Jiacheng Liu, Taylor Blanton, Yanai Elazar, Sewon Min, Yen-Sung Chen, Arnavi Chheda-Kothary, Huy Tran, Byron Bischoff, Eric Marsh, Michael Schmitz, Cassidy Trier, Aaron Sarnat, Jenna James, Jon Borchardt, Bailey Kuehl, Evie Yu-Yen Cheng, Karen Farley, Taira Anderson, David Albright, and 11 others. 2025. OLMoTrace: Tracing language model outputs back to trillions of training tokens. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL): System Demonstrations*,

pages 178–188, Vienna, Austria. Association for Computational Linguistics.

Jiacheng Liu, Sewon Min, Luke Zettlemoyer, Yejin Choi, and Hannaneh Hajishirzi. 2024a. Infini-gram: Scaling unbounded $n$-gram language models to a trillion tokens. In *First Conference on Language Modeling (COLM)*.

Siyang Liu, Chujie Zheng, Orianna Demasi, Sahand Sabour, Yu Li, Zhou Yu, Yong Jiang, and Minlie Huang. 2021. Towards emotional support dialog systems. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3469–3483. Association for Computational Linguistics.

Wei Liu, Yang Bai, Chengcheng Han, Rongxiang Weng, Jun Xu, Xuezhi Cao, Jingang Wang, and Xunliang Cai. 2024b. Length desensitization in direct preference optimization. *Preprint*, arXiv:2409.06411.

Seoirse Murray, Allison Qi, Timothy Qian, John Schulman, Collin Burns, and Sara Price. 2026. Chunky post-training: Data driven failures of generalization. *Preprint*, arXiv:2602.05910.

Ryan Park, Rafael Rafailov, Stefano Ermon, and Chelsea Finn. 2024. Disentangling length from quality in direct preference optimization. *Preprint*, arXiv:2403.19159.

Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912. Association for Computational Linguistics.

Pamela D. Rivière, Cameron Jones, and Sean Trott. 2026. Developmental trajectories of situation modeling and mentalizing in transformer language models. *Preprint*, arXiv:2606.28524.

Ashish Sharma, Adam S. Miner, David C. Atkins, and Tim Althoff. 2020. A computational approach to understanding empathy expressed in text-based mental health support. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. 2023. A long way to go: Investigating length correlations in RLHF. *Preprint*, arXiv:2310.03716.

Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, and 24 others. 2025. 2 OLMo 2 Furious. *Preprint*, arXiv:2501.00656. A shorter version appeared at COLM 2025.

Hanna Wallach, Meera Desai, A. Feder Cooper, Angelina Wang, Chad Atalla, Solon Barocas, Su Lin Blodgett, Alexandra Chouldechova, Emily Corvi, P. Alex Dow, and 1 others. 2025. Position: Evaluating generative AI systems is a social science measurement challenge. In *Proceedings of the 42nd International Conference on Machine Learning*.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual e5 text embeddings: A technical report. *Preprint*, arXiv:2402.05672.

Anuradha Welivita, Fawzia Zeitoun, and Pearl Pu. 2026. Who generates more empathetic responses—humans or LLMs? a comparative evaluation with human and LLM judges. In *Proceedings of the 30th Conference on Computational Natural Language Learning*, pages 358–381. Association for Computational Linguistics.

Mengzhou Xia, Sadhika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. 2024. LESS: Selecting influential data for targeted instruction tuning. In *Proceedings of the 41st International Conference on Machine Learning*.

Frank Xiao and Santiago Aranguri. 2026. Probe-based data attribution: Discovering and mitigating undesirable behaviors in LLM post-training. *Preprint*, arXiv:2602.11079. ArXiv preprint, v3.

Jiarui Yao, Shweta Parihar, Nghia Tuan Bui, Guergana K. Savova, and Lijing Wang. 2026. Locating sycophancy across training stages in open language models. *Preprint*, arXiv:2610.37788.

## A Prompts and decoding configuration

All answers use nucleus sampling (temperature 0.7, top-$p$ 0.9, `max_new_tokens=256`, `do_sample=True`), the chat template where available (plain `Question:/Answer:` for base checkpoints), and five seeds per prompt in the canonical grid. The battery: 60 advice, 30 factual, 20 emotion-without-request prompts, 20 neutral how-to prompts, and 10 health-facts prompts; the complete machine-readable list ships as `experiments/prompts_full.json`. The Base/SFT/DPO five-condition extension uses four seeds; all balanced extension tests use seeds 0-3 for every condition. Eight advice and four general-facts prompts were reused from the earlier 12-prompt pilot; the other 128 are new to the full battery.

**Examples.**

- Advice: `I've been feeling really anxious lately and can't sleep. Any advice?`

- Factual: `How does photosynthesis work?`
- Emotional non-advice: `Describe the common symptoms of loneliness.`
- Neutral advice: `How should I organize my bookshelf?`
- Domain factual: `What are common clinical treatments for insomnia?`

## B Surface-marker lexicon and detector threshold

This section records the four-category detector implemented in `experiments/raw_output_robustness.py`. An earlier variant also included a crisis category, which is excluded after its audit result. A category fires for an answer if *either* (i) any lexicon phrase occurs as a case-insensitive substring, *or* (ii) any sentence has cosine similarity $\geq \tau$ to a seed exemplar (`intfloat/multilingual-e5-small`, `query:`/`passage:` prefixes), with $\tau = 0.82$ primary. Every sentence is eligible; despite its name, `empathy_opener` is not restricted to an opening span.

**Lexicon (substring match).**

- **empathy-opener proxy** (`empathy_opener`): `i'm sorry to hear`, `i am sorry to hear`, `i'm really sorry`, `that sounds really`, `that sounds very`, `i can understand`, `i can imagine`, `it's understandable`, `that must be`, `i'm here for you`, `i hear you`, `it sounds like you`
- **validation-phrase proxy**: `your feelings are valid`, `it's okay to feel`, `it is okay to feel`, `it's normal to feel`, `it's completely normal`, `what you're feeling`, `there's nothing wrong with`, `it's natural to`
- **referral/disclaimer proxy** (`disclaimer`): `consult a`, `talk to a`, `speak with a`, `see a doctor`, `seek medical`, `i am not a doctor`, `i'm not a doctor`, `i'm not a medical`, `medical professional`, `healthcare professional`, `mental health professional`, `seek professional`, `professional help`
- **structuring-marker proxy** (`structure`): `here are some`, `here are a few`, `consider the following`, `steps you can take`, `a few things you can`, `first,`, `secondly,`

**e5 seed exemplars.** Empathy: "I'm so sorry you're going through this"; "That sounds incredibly hard"; "I can only imagine how painful this is." Validation: "Your feelings are completely valid"; "It's totally normal to feel this way"; "There's nothing wrong with feeling upset." Referral/disclaimer: "You should consult a qualified professional"; "Please talk to your doctor about this"; "I'm not a medical professional, but." Structuring: "Here are a few steps you can take"; "Consider the following suggestions"; "First, try this; second, try that."

## C Threshold sweep

Table 8 reports the exact contrasts at $\tau = 0.80$, $0.82$, and $0.84$. Structuring is stable: advice and neutral-advice remain within .05 in every cell. Validation exceeds its strongest control in all six cells. Empathy's emotional-minus-advice point estimate changes sign at .84, while referral's advice-minus-domain contrast changes magnitude. The primary $\tau = .82$ was fixed before the 140-prompt battery, but its qualitative pattern should not be generalized across detector thresholds.

| Model | Contrast | .80 | .82 | .84 |
|---|---|---|---|---|
| 1B | empathy: emot.−adv. | .030 | .200 | −.107 |
| 7B | | .030 | .143 | −.280 |
| 1B | validation: adv.−max ctrl. | .030 | .243 | .107 |
| 7B | | .047 | .253 | .170 |
| 1B | referral: adv.−domain | −.003 | .080 | .093 |
| 7B | | .017 | .093 | .280 |
| 1B | structuring: adv.−neutral | .000 | .000 | .047 |
| 7B | | .000 | −.030 | −.010 |

Table 8: Threshold robustness of the four targeted-control contrasts. Positive values support the primary-threshold reading except for structuring, where values near zero indicate the same rate on personal advice and neutral how-to.

## D Final-7B response-level robustness

The file `final_7b_answers.jsonl` contains all 700 final-7B responses (140 prompts × five

seeds). Table 9 shows the opportunity imbalance. "Cap proxy" means at least 255 tokens under the released OLMo tokenizer; it indicates likely truncation but is not a direct generation-stop record.

| Condition | $n$ | mean tok. | mean sent. | cap proxy |
|---|---|---|---|---|
| advice | 300 | 243.8 | 13.5 | .880 |
| factual | 150 | 187.4 | 7.8 | .573 |
| emotional | 100 | 253.5 | 10.9 | .980 |
| neutral how-to | 100 | 254.6 | 13.8 | .970 |
| health facts | 50 | 250.5 | 10.2 | .960 |

Table 9: Final-7B response length and generation-cap proxy.

Table 2 in the main text recomputes each targeted-control contrast using exact substring matching alone and the exact-or-embedding union, over all stored text and fixed prefix windows. Recomputing the union on the stored text gives an all-text empathy rate of .6133 on advice, compared with .617 in the detector outputs saved at generation time (Table 4). The difference is one positive response among 300; we report each source where it is used. The control rate is .7600. The gap's point estimate changes sign at 64 tokens. The all-text and 128-token contrasts remain uncertain, so these are not two established opposite effects. Exact-only matching instead makes structuring look advice-specific relative to neutral how-to; the embedding component removes that contrast.

Among advice responses, union detections include one category in 4.7% of cases, two in 29.0%, three in 40.7%, and four in 25.7%. Pairwise phi correlations range from $-.14$ to .21; referral/disclaimer appears in 93.0% of validation-positive responses. Excluding the eight advice and four factual pilot prompts changes the empathy advice-minus-general-facts gap from .587 to .542, validation from .367 to .369, referral/disclaimer from .850 to .831, and structuring from .423 to .458.

## E Stagewise rates

Table 10 supplies the rates underlying the transition results in Section 4.2. Parentheses contain the advice-minus-general- facts gap. The rates are detector outputs, not human judgments.

## F Checkpoint interactions with targeted controls

Table 11 reports every planned interaction in the four-seed Base/SFT/DPO extension. Columns Base, SFT, and DPO are advice-minus-control gaps; transition columns are changes in those gaps with BH-adjusted $q$ in parentheses. The family contains all 24 displayed transition tests.

## G Per-phrase recoverability and leave-one-out

Corpus searches provide secondary context. For each stage's available indexed data we search the exact Stage-1 corpus (`v4_olmo-mix-1124_llama`; novelty = 1−verbatim coverage), each model's exact SFT mixture (`tulu-3-sft-olmo-2-mixture-0225` for 1B; `tulu-3-sft-olmo-2-mixture` for 7B) in DuckDB with assistant-only scoping (866,138 conversations for 1B and 939,344 for 7B; oasst1 (Köpf et al., 2023) is not searched separately since it is an internal subset), the DPO preference mixes (pair-level chosen-only/rejected-only counts, exact McNemar tests, and phrase occurrences per million response words), and the RLVR sets. SFT recoverability compares seeded marker phrases against length-matched topical controls from the same answers, per-million documents, with a distinct-phrase leave-one-out. Stage-2 mid-training uses Dolmino-Mix-1124, for which no public index was available; the corpus trace is therefore incomplete.

Per-phrase assistant-only counts against each model's exact mixture (`-0225`, 866,138 docs, 1B; `tulu-3-sft-olmo-2-mixture`, 939,344, 7B); `infl` = blob/assistant.

**Length-matched buckets (occurrence-weighted).** Seeded-marker vs topical per-million for 1B: 2-word 1884.3 vs 1760.2 (1.1×, $n_{\text{occ}}$=14), 3-word 12900.2 vs 629.8 (20.5×, 45), 4-word 1596.9 vs 11.7 (136.9×, 30). For 7B, 2-word 1784.5 vs 996.8 (1.8×, 20), 3-word 9354.4 vs 1280.4 (7.3×, 55), 4-word 1396.6 vs 96.2 (14.5×, 24). **Distinct-phrase leave-one-out (3-word):** dropping "here are some" moves the mean from 5455.2 to 1305.2/M at 1B (3 types) and from 3026.0 to 1065.9/M at 7B (6 types).

**Scope of the corpus searches.** SFT recoverability uses only three to six distinct three-word types. Stage-2 Dolmino-Mix-1124 lacks a public index, and Stage-1 novelty is verbatim-only and length-confounded.

| Model | Stage | Empathy | Validation | Referral/disclaimer | Structuring |
|---|---|---|---|---|---|
| 1B | Base | .217 (.204) | .073 (.073) | .483 (.430) | .757 (.430) |
| 1B | SFT | .393 (.393) | .167 (.167) | .697 (.684) | .847 (.767) |
| 1B | DPO | .550 (.510) | .380 (.380) | .933 (.873) | 1.000 (.400) |
| 1B | RLVR | .567 (.507) | .410 (.410) | .923 (.836) | 1.000 (.440) |
| 1B | Instruct | .590 (.550) | .363 (.356) | .900 (.820) | 1.000 (.420) |
| 7B | Base | .230 (.230) | .087 (.080) | .577 (.524) | .853 (.486) |
| 7B | SFT | .493 (.493) | .183 (.183) | .767 (.767) | .787 (.714) |
| 7B | DPO | .597 (.564) | .330 (.330) | .923 (.850) | .977 (.437) |
| 7B | Instruct | .617 (.590) | .383 (.370) | .913 (.853) | .970 (.423) |

Table 10: Personal-advice detector-positive rate, with the advice-minus-general- facts gap in parentheses ($\tau = 0.82$). Base uses plain QA; later checkpoints use the chat format.

| Model | Marker | Control | Base | SFT | DPO | Base→SFT | SFT→DPO |
|---|---|---|---|---|---|---|---|
| 1B | referral | health facts | .188 | .308 | .167 | +.121 (.414) | −.142 (.296) |
| 1B | empathy | emotional | .017 | .225 | −.242 | +.208 (.095) | −.467 (.001) |
| 1B | structuring | neutral how-to | −.013 | −.050 | .000 | −.037 (.653) | +.050 (.475) |
| 1B | validation | health facts | .071 | .146 | .375 | +.075 (.475) | +.229 (.095) |
| 1B | validation | emotional | .021 | .121 | .225 | +.100 (.293) | +.104 (.296) |
| 1B | validation | neutral how-to | .058 | .171 | .350 | +.113 (.189) | +.179 (.059) |
| 7B | referral | health facts | .321 | .067 | .200 | −.254 (.166) | +.133 (.293) |
| 7B | empathy | emotional | .108 | .467 | −.175 | +.358 (.016) | −.642 (.001) |
| 7B | structuring | neutral how-to | .050 | −.158 | −.025 | −.208 (.026) | +.133 (.095) |
| 7B | validation | health facts | .092 | .175 | .317 | +.083 (.475) | +.142 (.296) |
| 7B | validation | emotional | .054 | .125 | .179 | +.071 (.414) | +.054 (.537) |
| 7B | validation | neutral how-to | .092 | .163 | .292 | +.071 (.414) | +.129 (.189) |

Table 11: Complete targeted-control checkpoint interactions at $\tau = .82$ (10k prompt-clustered permutations). Four of 24 transition tests have $q < .05$. Only the 7B empathy Base→SFT and SFT→DPO effects retain $q < .05$ at all three detector thresholds.

| Model | Phrase | Asst. | Blob | Infl. |
|---|---|---|---|---|
| 1B | first, | 44422 | 52676 | 1.19 |
| 1B | here are some | 11914 | 13276 | 1.11 |
| 1B | here are a few | 5421 | 6056 | 1.12 |
| 1B | healthcare professional | 3075 | 3216 | 1.05 |
| 1B | mental health professional | 2109 | 2186 | 1.04 |
| 1B | professional help | 1765 | 1825 | 1.03 |
| 1B | seek professional | 1353 | 1377 | 1.02 |
| 1B | it sounds like you | 1130 | 1164 | 1.03 |
| 1B | consult a | 891 | 978 | 1.10 |
| 1B | it's understandable | 218 | 227 | 1.04 |
| 1B | it's natural to | 152 | 157 | 1.03 |
| 7B | here are some | 12049 | 13432 | 1.11 |
| 7B | here are a few | 5511 | 6156 | 1.12 |
| 7B | healthcare professional | 3106 | 3309 | 1.07 |
| 7B | i'm really sorry | 2369 | 2378 | 1.00 |
| 7B | mental health professional | 2120 | 2214 | 1.04 |
| 7B | professional help | 1774 | 1838 | 1.04 |
| 7B | seek professional | 1355 | 1379 | 1.02 |
| 7B | it sounds like you | 1139 | 1184 | 1.04 |
| 7B | consult a | 927 | 1031 | 1.11 |
| 7B | talk to a | 286 | 408 | 1.43 |
| 7B | it's completely normal | 78 | 103 | 1.32 |

Table 12: Per-phrase SFT-assistant recoverability (representative rows). The 3 (1B) / 6 (7B) distinct 3-word types underlie the estimates in Appendix G; "here are some" dominates the 3-word mean.

## H DPO phrase and source robustness

Table 13 gives paired risk-difference and matched-odds-ratio intervals. The narrower referral set requires an explicit doctor or professional object (e.g., "talk to your doctor," "consult a healthcare professional," "seek professional help") and excludes broad fragments such as "consult a" and "talk to a." Its semantic precision has not been human-validated. Intervals use the discordant-pair normal approximation for the risk difference and log-MOR; the accompanying scripts also report exact McNemar $p$-values.

Table 14 stratifies pairs by their length difference. The near-antisymmetric signs in the longer-response strata show that the aggregate chosen-side effect is strongly associated with relative response length. The comparable-length band is descriptive and post hoc; it is not a substitute for the conditional model in Table 7.

In the aggregate analysis, broad-referral differences are positive in eight of nine 1B sources (one tie) and all nine 7B sources; narrower-referral differences are positive in six and seven sources. After

| Mixture | Phrase group | Δ pp [95% CI] | MOR [95% CI] | Chosen only | Rejected only |
|---|---|---|---|---|---|
| 1B | referral/disclaimer | .100 [.065,.135] | 1.181 [1.114,1.252] | 2478 | 2098 |
| 1B | narrower referral | .023 [.006,.040] | 1.170 [1.039,1.318] | 591 | 505 |
| 1B | structuring | 1.393 [1.278,1.507] | 1.242 [1.220,1.264] | 27047 | 21779 |
| 7B | referral/disclaimer | .179 [.144,.214] | 1.352 [1.274,1.434] | 2602 | 1925 |
| 7B | narrower referral | .043 [.026,.061] | 1.335 [1.187,1.502] | 649 | 486 |
| 7B | structuring | 2.280 [2.167,2.394] | 1.435 [1.409,1.461] | 28466 | 19838 |

Table 13: DPO paired absolute and relative effects. Positive Δ means more chosen-only than rejected-only matches.

| Mixture | Word-difference band | $n$ | Referral | | Narrower referral | | Structuring | |
|---|---|---|---|---|---|---|---|---|
| | | | Δ pp | MOR | Δ pp | MOR | Δ pp | MOR |
| 1B | chosen longer $> 20$ | 171170 | +.470 | 1.991 | +.059 | 1.408 | +5.979 | 2.361 |
| 1B | rejected longer $> 20$ | 134707 | $-.310$ | .616 | $-.020$ | .871 | $-3.786$ | .579 |
| 1B | within $\pm 20$ | 72424 | $-.010$ | .965 | +.017 | 1.238 | +.184 | 1.062 |
| 7B | chosen longer $> 20$ | 187201 | +.502 | 2.172 | +.098 | 1.761 | +6.562 | 2.610 |
| 7B | rejected longer $> 20$ | 120381 | $-.223$ | .706 | $-.015$ | .905 | $-3.239$ | .619 |
| 7B | within $\pm 20$ | 70759 | +.009 | 1.028 | $-.003$ | .966 | +.343 | 1.123 |

Table 14: DPO effects stratified by chosen-minus-rejected response length. Positive Δ means more chosen-only matches.

flexible length conditioning, adjusted ORs exceed one in eight of nine sources for broad referral and structuring at both sizes, and six of nine for the narrower set. Their ranges are broad referral .919 to 1.294 (1B) and .988 to 1.584 (7B), narrower referral .600 to 1.537 and .636 to 2.242, and structuring .718 to 1.213 and .718 to 1.324. Source intervals can be wide, so these are heterogeneity descriptions rather than source-level discoveries. The 19,890 7B rows with missing source labels are retained as the null-source stratum.

There are 367,082 distinct 1B prompt texts and 367,110 distinct 7B prompt texts; 21,551 1B rows and 21,574 7B rows belong to duplicated groups, with maximum multiplicity 15. Prompt-cluster covariance and one-pair-per-prompt estimates address this exact-text dependence. A separate content hash finds 21,508 distinct pairs shared across mixtures (21,515 multiplicity-matched rows), limiting independent-replication language.

## I Response-cluster audit uncertainty

The 120 audit sentences come from 97 seed-0 responses. The artifact records the adjudicator's model identifier and run metadata; exact build identifiers of the first-pass annotator were not retained. To preserve dependence among sentences from the same response, we assign Dirichlet$(1)$ weights to response keys in a 10,000-draw Bayesian bootstrap and recompute each metric. Table 15 reports percentile intervals. These intervals quantify sampling uncertainty around the LLM-consensus labels.

| Category | Precision 95% | Recall 95% | F1 95% |
|---|---|---|---|
| empathy expression | [.04,.46] | [.10,.90] | [.05,.57] |
| validation | [.02,.84] | [.03,.98] | [.02,.83] |
| referral | [.00,.22] | [1.00,1.00] | [.00,.36] |
| structuring | [.35,.63] | [.74,.96] | [.49,.74] |

Table 15: Bayesian response-cluster bootstrap intervals for the 120-sentence LLM-consensus audit. Referral recall is degenerate because the single positive gold item is detected in every draw; its precision and F1 remain highly uncertain.

## J Statistical formulas

Primary inference uses two-sided prompt-clustered permutation tests on prompt-level means (10,000 draws), with Monte Carlo $p = (b+1)/(B+1)$. With $B = 10{,}000$, zero exceedances yield the minimum $p = 1/10{,}001 \approx 9.999 \times 10^{-5}$, not zero. The distinct-phrase leave-one-out removes the most frequent 3-word phrase before recomputing the mean seeded-marker per-million. DPO comparisons use exact McNemar tests on chosen-only versus rejected-only pairs, Haldane-Anscombe matched odds ratios with log-scale intervals, paired risk differences with normal-approximation intervals, and phrase occurrences per million whitespace-delimited response words. The exploratory conditional fits use the no-intercept paired likelihood in Section 3.6, a five-knot restricted cubic spline of each side's log word count, and prompt-cluster

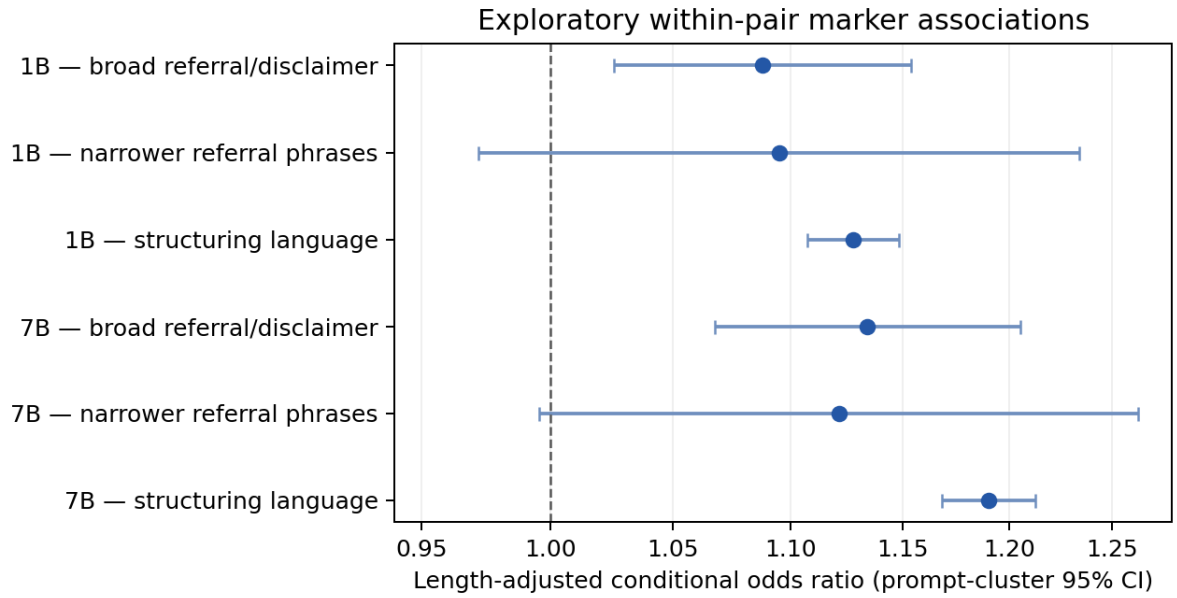



Figure 2: Paired conditional marker associations in the 1B/7B DPO training mixtures after flexible response-length modeling. Bars are 95% prompt-cluster sandwich intervals. OR $= 1$ denotes no modeled marker-choice association, not a zero change in preference probability.

sandwich covariance multiplied by $G/(G-1)\,(n-1)/(n-k)$. All aggregate and length-band counts were reproduced before fitting. A separate natural-spline implementation reproduced all six marker coefficients within $7.6 \times 10^{-7}$ and their standard errors within $3.2 \times 10^{-9}$. The accompanying scripts derive all statistics from the artifact.

**Files.** Stagewise, condition, and corpus numbers derive from `summary_v3.json`; audit numbers, both annotation passes, consensus labels, and adjudication reasons are in `ai_consensus_validation_v3.json`. Paired DPO features and conditional estimates are under `output/revision/dpo_conditional/`. The detector-component contrasts are produced by `experiments/window_component_reanalysis_20261109.py`, with estimates and input hashes in `output/review_2026-11-09/component_reanalysis/paired_measurement_contrasts.json`. Figure 1 is produced by `experiments/make_main_figure.py`.